

**CONWAY REGIONAL HEALTH SYSTEM
WOMENS' AND INFANTS' SERVICES**

TITLE: GLUCOSE MANAGEMENT PROTOCOL: NEWBORN

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POLICY #: G3

EFFECTIVE DATE: 2/09

**REVISION DATE(S): 12/13, 11/17, 6/18,
04/21, 07/23, 4/24**

POLICY

To provide guidelines to be followed in monitoring neonates who may be at risk for unstable glucose. According to The American Academy of Pediatrics, Committee on the Fetus and Newborn, the definition of hypoglycemia is a value less than 45 mg/dl for symptomatic infants and less than 40 for asymptomatic infants.

DEFINITIONS

LGA: Large for gestational age is defined as 4000 g or 90th percentile or above.

AGA: Appropriate for gestational age is defined as 2500g to 4000g.

SGA: Small for gestational age is defined as below 2500 g or 10th percentile or below.

Macrosomia is defined as >4500g per The American College of Obstetricians and Gynecologists (ACOG).

IUGR: Intrauterine growth restriction or fetal growth restriction.

PREVENTION OF HYPOGLYCEMIA

Efforts to help prevent neonatal hypoglycemia should be initiated following birth to include:

- Skin to Skin after delivery. Cover mom/baby dyad for increased heat conservation
- Newborn temperature monitoring and thermoregulation of infant and environment.
- Early feed during neonatal transition period; if breastfeeding initiate nursing opportunity within the first 30 – 60 minutes of birth.
- For at risk newborns encourage regular and frequent feeds.

NEWBORN GLUCOSE MANAGEMENT PROTOCOL

1. Glucose screen to be performed on newborns within 30 minutes of feeding when meeting the following criteria: SGA, LGA, IUGR, late preterm (34-36 weeks gestation), or mother with history of GDM/Diabetes. See table on last page.
2. Glucose screen immediately upon recognition of any of the signs or symptoms of hypoglycemia such as those listed on the clinical pathway (last page).
3. If infant is greater than 35 weeks gestation and meets criteria for hypoglycemia symptoms, implement **glucose gel protocol** according to the algorithm and by

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infant's weight. The glucose gel will be supplied by Pharmacy in 2ml prefilled syringes.

Weight in kilograms	Dosage amount
2 kg	1 ml
2.5 kg	1.25 ml
3 kg	1.5 ml
3.5 kg	1.75 ml
4 kg	2 ml
4.5 kg	2.25 ml
5 kg	2.5 ml

- The gel will be administered by the nurse to the infant by applying it to the buccal mucosa of each cheek in 0.5 ml increments, massaging into the gums and cheek as tolerated, by a gloved finger, alternating sides until the total dose is received. Follow clinical pathway on last page.
- May repeat glucose gel administration up to a total of three gel administrations, if indicated per the Clinical Pathway for Glucose Management algorithm. If glucose level remains less than 40 after second glucose gel administration and glucose check and infant requires a third dose of gel then notify Pediatrician and initiate IV glucose therapy per the Clinical Pathway/Protocol. No more than three doses of glucose gel during the first 48 hours.
- Infants with NPO orders will continue to have their glucose checked every 6 hours once glucose levels have been stabilized.
- Glucose gel should only be administered for the treatment of hypoglycemia of asymptomatic newborns and as outlined in the Clinical Pathways for Glucose Management in the Newborn.

PROCEDURE – GLUCOSE SPECIMEN COLLECTION

- Explain procedure to parents.**
- Wash hands and don gloves.
- Place infant in secure location.
- Locate area for heelstick, swab with alcohol and allow to air dry.

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5. Perform heelstick using approved lancet.
6. Remove lancet, wipe first drop gently with gauze.
7. Immediately process the specimen on the Nursery Accu Check glucose monitor, ensuring correct patient information is included on log.
8. Apply pressure to sight with gauze and/or adhesive bandage
9. Document the glucose level. Perform venipuncture to obtain serum glucose when heelstick result less than 40.

Subsequent monitoring and titration of IV Therapy

- Glucose IV therapy can be initiated by the following: Administer 2mL/kg D10W IVP and start continuous infusion of D10W at 3.3mL/kg/hr (80mL/kg/day).
- Recheck blood glucose in 15-30 minutes from time of bolus administration.
- Plasma glucose concentration should be measured 30 to 45 minutes after the initiation of IV therapy, and the infusion rate or dextrose concentration adjusted as needed to maintain BG greater than 45 mg/dL during the first 48 hours with an upper limit of 90 to 100 mg/dL.
- Repeat BG levels should be obtained 30 to 45 minutes after any change in the IV dextrose infusion rate.
- Maximum infusion rate: The maximum rate of glucose infusion for treatment is limited by the maximum amount of fluids that can be administered to patient. If the glucose infusion rate exceeds 12 mg/kg per minute or infusion rates exceeds 160 ml/kg per day, notify pediatrician and receive order for D12.5.
- Patient's fluid balance and clinical status should be monitored closely for volume overload.
- Weaning and transition to oral feeds: When the BG is stabilized and maintained within the target range, the glucose infusion rate can be tapered slowly, and the infant can be transitioned to oral feeds.
- Weaning is started when the infant's BG levels have been in the target range for six to nine hours.
- If BG levels are above the target range, the glucose infusion rate should be decreased sooner.
- Fluids should be decreased by 2mL/hr and glucose rechecked in 3 hours, if BG level stays above target range continue titration process decreasing fluids by 2ml/hr. Do not titrate fluids if BG is less than or equal to 45 mg/dL.

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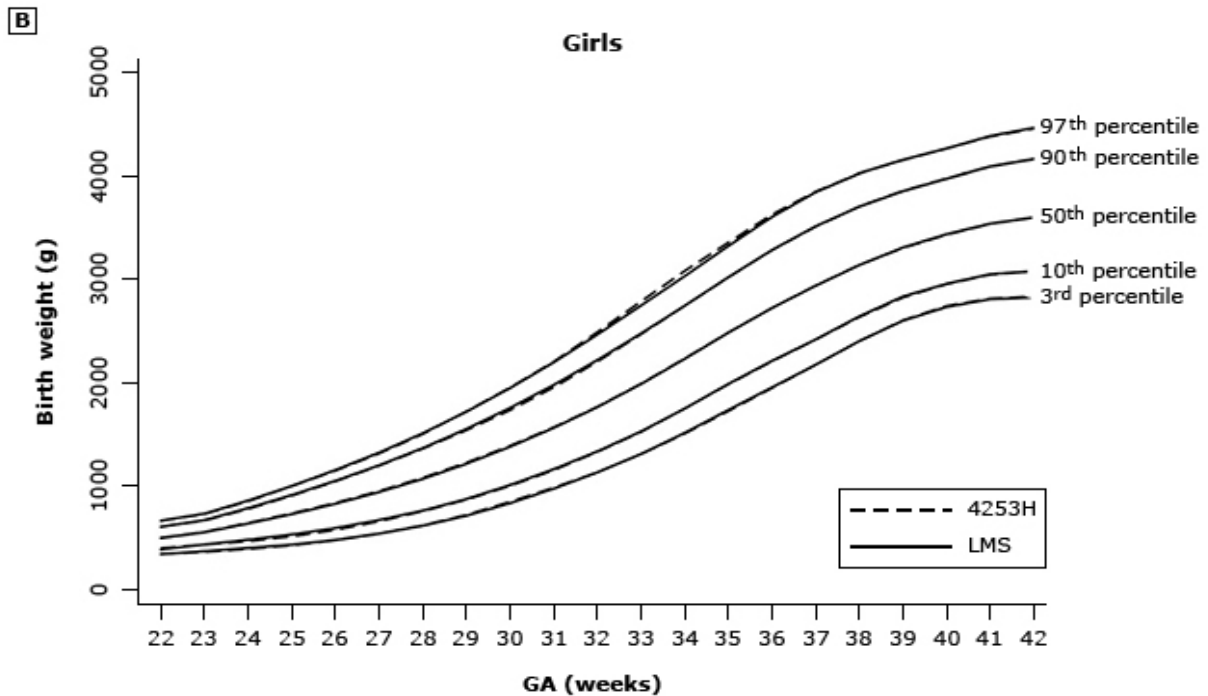
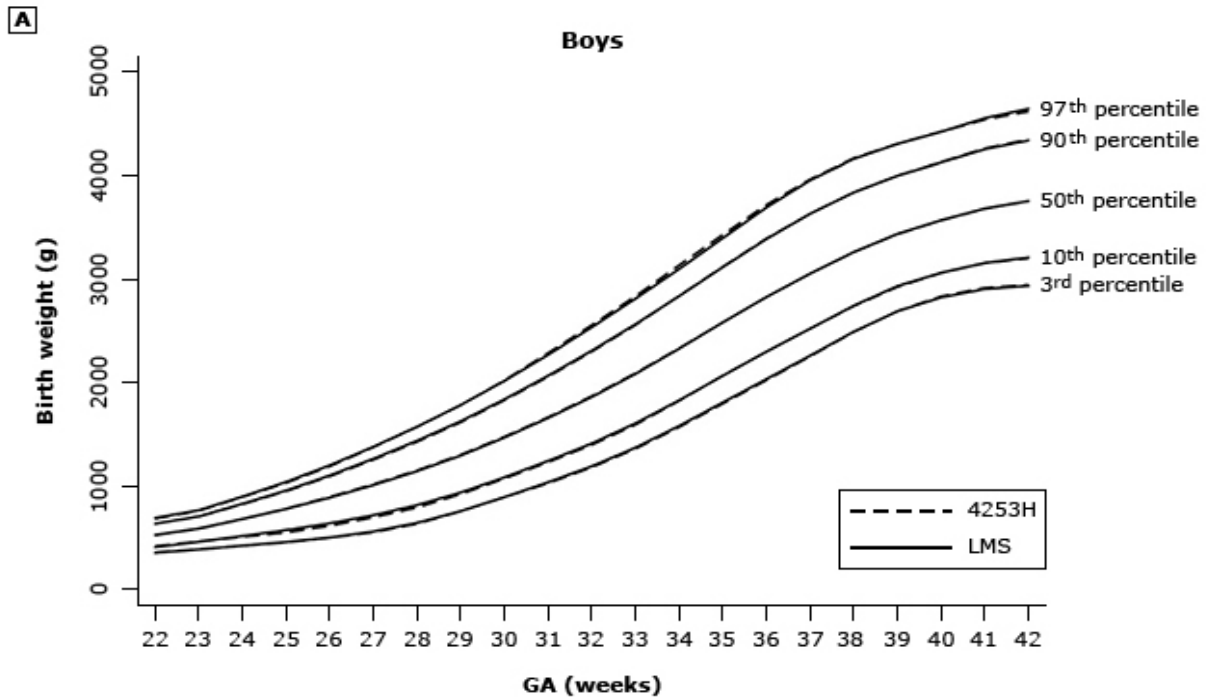
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<https://www.chop.edu/clinical-pathway/glucose-monitoring-healthy-newborn-clinical-pathway>

Clinical Pathways for Glucose Management in the Newborn

Neonatal Glucose Monitoring

Assess for symptoms of hypoglycemia:
jitteriness, irritability, high-pitched cry, apnea, cyanosis, irregular/rapid breathing,
hypotonia, seizures, temperature instability, hypothermia, poor suck, refusal to eat

Assess for risk factors for hypoglycemia:
SGA, LGA, IUGR, late preterm (35-36.6 weeks), infant of diabetic mother (GDM or DM I/DM II)

Asymptomatic
Without risk factors

No further action

BG greater than 40
Check BG pre-feed (Q2-3 hour) for 2 more consecutive feeds.

May stop checking BG when greater than 40 for less than 24 hours of age; or greater than 50 for 24-48 hours of age.

Asymptomatic **with** risk factors

Feed within 1 hour of life and
Check blood glucose (BG) 30 minutes after feed

BG 25-40

Apply gel and refeed
Recheck BG 1 hour after gel

May gel up to 3 separate doses,
notify provider to implement
IV dextrose treatment

BG less than 25

Apply gel and refeed
Recheck BG 1 hour after gel

Notify provider for
management plan;
consider IV dextrose treatment

Symptomatic

Check blood glucose (BG)

BG greater than 45

Notify provider of
symptoms

BG 45 or less

notify provider for
management plan
consider IV dextrose
treatment